

Shayan Ghiasvand

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📍 Toronto, ON

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SKILLS

- **Programming:** Python, C/C++, MATLAB, Git
- **Libraries:** Tensorflow, Keras, Pytorch, Scikit-learn, OpenCV, Open3D, Scikit-image
- **Engineering:** Simulink, PLC, ROS, Arduino, Microcontroller, Kalman Filter
- **CAD:** SOLIDWORKS, AutoCAD, PTC Creo, 3D Print
- **Robotic Simulation:** RoboDK, Gazebo, CARLA
- **Other:** GNU/Linux, Embedded Systems, Communication Protocol, UDP, UART, SSP, I^2C , PWM

EXPERIENCE

- **University of Toronto** Toronto, Canada
Jul 2024 - Oct 2024
Robotics Engineer
Robotized an RC excavator by modifying the electrical circuit of the hydraulic system and integrating an **Arduino**. ([video](#))
Enabled **wireless communication** for accurate pose commands using a **bluetooth module**.
- **Concordia University** Montréal, Canada
Sep 2021 - Dec 2023
Research Assistant, AI Robotics
Vision-based Object Tracking with two data-driven methods:
 - **Deep Learning:** Introduced an original **real-time** vision-based pose estimation method, demonstrating a **32% reduction in pose error** and a **22% improvement in velocity precision** compared to conventional methods. ([video](#))
 - **Image Feature Optimization:** Optimized a novel image feature function to reduce motion coupling in visual servoing, achieving an **87% improvement in decoupling efficiency**.**Novel Imagery Dataset Generation:** Specific for pose estimation, created a benchmark dataset of 600K+ synthetic and real data. Employed **DBSCAN** for outlier removal. ([video](#))
- **RoboDK** Montréal, Canada
May 2022 - Sep 2022
Intern, Robotic Software Developer
Developed RoboDK software Add-ins:
 - **Real-time IBVS:** Segmenting, and tracking 2D markers using a robot, providing an automation add-in to the software.
 - **3D Reconstruction** by **Point Cloud Registration**Developed an automatic **bin-filling algorithm** as a base for **object pose estimation** and **bin-picking** algorithms. ([video](#))
- **Chemnitz University of Technology** Chemnitz, Germany
May 2019 - Jul 2019
Intern, Mechanical Design Engineer
Tuned and calibrated the **PID controller** of the tensile fatigue test equipment, significantly improving accuracy in load control and responsiveness. Introduced a novel method for the real-time monitoring of the tensile fatigue test by applying the **Acoustic Emission Sensor** and collecting data. Conducted FEM Analysis of Shaft-Hub Connections, identifying stress distribution and potential failure points. Performed 3D Microscope Imaging and Analysis of crack growth.

PROJECTS

- **Simulated Robotic Combat: Path Planning**
Developed a novel **path planning** algorithm in an obstructed environment, demonstrated **faster performance** than established methods. **Implemented an image processing algorithm**, accurately detecting the robots and the obstacles. Specified **dynamic target points** for attacker and defender robots to **optimize combat strategies**.
- **Ball & Beam Controller Design & GUI**
Improving the **stability** of the ball & beam by implementing linear and sliding mode controllers. Designed a **MATLAB GUI** for the ball & beam system, allowing users to set control parameters and initial conditions to **animate the experiment**, enabling **method comparison** and easy **tuning**.
- **Pick-and-Place to Recycle Objects**
Automating recycling operation using a robotic manipulator in a Gazebo environment. Implementing a **pick-and-place algorithm** by employing **point cloud processing** for object detection and applying **RRT path planning** with real-time **ROS** integration to achieve collision-free trajectories.

EDUCATION

- **Concordia University** Montréal, Canada
Master of Applied Science in Mechanical Engineering
May 2024
Courses: Autonomous Robotics Systems — Industrial Automation — Industrial Control Systems — Adaptive Control
- **University of Tehran** Tehran, Iran
Bachelor of Science in Mechanical Engineering
May 2019
Courses: Mechatronics — Automatic Control

PUBLICATIONS

- **Ghiasvand, S., Xie, W.-F., & Mohebbi, A. (2024).** "Deep neural network-based robotic visual servoing for satellite target tracking." *Frontiers in Robotics and AI*